

Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

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Chapter 1

Basic Setup

Given a "high dimensional" data set $\{\mathbf{x}_i\}_{i=1}^n$ where $\mathbf{x}_i \in \mathbb{R}^p$ for all $i = 1, \dots, n$, the statistician fixes a positive integer $q < p$ and wishes to find the q principal components. Assuming that the data is centred (that is, with mean $\mathbf{0}$), the probabilistic principal component analysis model is a statistical model that treats $\mathbf{x}_i$'s as realizations of independent random variables $\mathbf{X}_i = \mathbf{W}\mathbf{Z}_i + \boldsymbol{\epsilon}_i$, where $\mathbf{W}$ is an unknown $p \times q$ loading matrix, $\mathbf{Z}_i \sim N_q(\mathbf{0}, \mathbf{I}_q)$ is the latent normal variable, and $\boldsymbol{\epsilon}_i \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I}_p)$ is the Gaussian noise with unknown standard deviation $\sigma > 0$, and $(\mathbf{Z}_1, \dots, \mathbf{Z}_n, \boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_n)$ are all independent (for now). Put $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$ for the unknown parameter to estimate and $p(\mathbf{x}; \theta)$ for the density of $\mathbf{X}_1$, namely the density of $N_p(\mathbf{0}, \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 \mathbf{I}_p)$.

We now setup the quotient space structure. Write Θ for the collection of all tuples $(\mathbf{W}, \sigma^2)$ such that $\mathbf{W}$ is a real $p \times q$ matrix with $\text{rank}(\mathbf{W}) = q$ and $\sigma^2 \in (0, \infty)$. Given that the collection of all real $p \times q$ matrix with rank q is an open subset of the Euclidean space $\mathbb{R}^{pq}$, we consider Θ as an open subset of the Euclidean space $\mathbb{R}^{pq+1}$, hence a Riemannian manifold with the standard Euclidean metric

$$g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)) = \text{tr}(\mathbf{V}\mathbf{U}^T) + cb, \quad (1.1)$$

whenever $(\mathbf{A}, s) \in \Theta$, $(\mathbf{V}, c), (\mathbf{U}, b) \in T_{(\mathbf{A}, s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$. Next, we define $(\mathbf{A}, s)$ and $(\mathbf{B}, t) \in \Theta$ to be equivalent when they give the same density, namely when $\mathbf{A}\mathbf{A}^T + s\mathbf{I}_p = \mathbf{B}\mathbf{B}^T + t\mathbf{I}_p$, and write $(\mathbf{A}, s) \sim (\mathbf{B}, t)$. The equivalence classes are denoted by $[(\mathbf{A}, s)]$ or more briefly $[\mathbf{A}, s]$. Let us denote by $O(q)$ the collection of all $q \times q$ real orthogonal matrices. It is shown in Datta's thesis that $[\mathbf{A}, s] = \{(\mathbf{A}\mathbf{Q}, s) \mid \mathbf{Q} \in O(q)\}$. Therefore, $M := \Theta / \sim$ can be identified as the orbits of the right action of $O(q)$ on Θ and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see [Lee(2013)], Theorem 21.10) shows that there is a unique smooth structure making M a manifold of dimension $pq + 1 - \frac{q(q-1)}{2}$ and $\pi : \Theta \rightarrow M$ with $\pi((\mathbf{A}, s)) = [\mathbf{A}, s]$ is a smooth submersion. Moreover, this action is verticle, transitive on fibers and is isomrtic with respect to (1.1). Thus, there exists a unique metric g on M making (M, g) a Riemannian manifold and π a Riemannian submersion (see [Lee(2018)], Theorem 2.28), meaning that

$$g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)) = g_{[\mathbf{A}, s]}(d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c)), d\pi_{(\mathbf{A}, s)}((\mathbf{U}, b))), \quad (1.2)$$

whenever $(\mathbf{A}, s) \in \Theta$, $(\mathbf{V}, c), (\mathbf{U}, b) \in T_{(\mathbf{A}, s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$.

1.1 Exponential Map and Logarithmic Map Computation

In this section, we carry out some basic computations on (M, g) . Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and verticle tangent spaces. We begin with a few technical details in the computation.

Proposition 1. Fix $(\mathbf{A}, s) \in \Theta$. The verticle tangent space $V_{(\mathbf{A}, s)} := \ker(d\pi_{(\mathbf{A}, s)})$ equals to the collection of all tangent vectors of the form $(\mathbf{A}\boldsymbol{\Omega}, 0) \in T_{(\mathbf{A}, s)}\Theta$, with $\boldsymbol{\Omega}$ ranged over all $q \times q$ skew symmetric matrices. The

horizontal tangent space $H_{(\mathbf{A}, s)} := V_{(\mathbf{A}, s)}^\perp$ equals to the collection of all tangent vectors of the form $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$ ranged over all $c \in \mathbb{R}$ and all $q \times q$ matrices $\mathbf{V}$ satisfying $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$.

Proof. We start with computing the verticle tangent space. Define function $\Phi : \Theta \rightarrow \mathbb{R}^{p \times p}$ by setting, for every $(\mathbf{A}, s) \in \Theta$:

$$\Phi((\mathbf{A}, s)) = \mathbf{A}\mathbf{A}^T + s\mathbf{I}_p. \quad (1.3)$$

We claim that

$$\ker(d\pi_{(\mathbf{A}, s)}) = \ker(d\Phi_{(\mathbf{A}, s)}).$$

Before proving the claim, let us first study $\ker(d\Phi_{(\mathbf{A}, s)})$. For $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$, define a curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ by setting $\gamma(t) = (\mathbf{A} + t\mathbf{V}, s + tc)$. Then:

$$d\Phi_{(\mathbf{A}, s)}((\mathbf{V}, c)) = \frac{d}{dt} \Big|_{t=0} \Phi(\gamma(t)) = \mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + 2sc\mathbf{I}_p.$$

If $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A}, s)})$, then

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + 2sc\mathbf{I}_p = \mathbf{0}. \quad (1.4)$$

If $\mathbf{u} \in \mathbb{R}^p$ is any nonzero vector orthogonal to all column vectors of $\mathbf{A}$, then $\mathbf{A}^T \mathbf{u} = \mathbf{0}$ and

$$0 = \mathbf{u}^T (\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + 2sc\mathbf{I}_p) \mathbf{u} = 2sc\mathbf{u}^T \mathbf{u}.$$

Given that $s > 0$, we must have $c = 0$ and (1.4) reduces to

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T = \mathbf{0}. \quad (1.5)$$

Apply $\mathbf{u}$ to the right to get:

$$\mathbf{A}\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Given that $\mathbf{A}$ has full rank, rank nullity theorem forces

$$\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Therefore, we have proven that every vector $\mathbf{u}$ orthogonal to columns vectors of $\mathbf{A}$ will also be orthogonal to column vectors of $\mathbf{V}$. This means $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ for some $q \times q$ matrix $\mathbf{\Omega}$. Substitute this back to (1.5) gives

$$\mathbf{A}(\mathbf{\Omega}^T + \mathbf{\Omega})\mathbf{A}^T = \mathbf{0}.$$

Applying $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ to the left and $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$ to the right yields $\mathbf{\Omega}^T + \mathbf{\Omega} = \mathbf{0}$. Therefore, $\mathbf{\Omega}$ is skew-symmetric. Conversely, (1.5) is satisfied by $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ with any skew-symmetric matrix $\mathbf{\Omega}$. We have proven that $\ker(d\Phi_{(\mathbf{A}, s)})$ equals to the collection of all tangent vectors of the form $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A}, s)}\Theta$, with $\mathbf{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

We now move back to prove the claim. Let us define function $F : M \rightarrow \mathbb{R}^{p \times p}$ by setting $F([\mathbf{A}, s]) = \Phi((\mathbf{A}, s))$, where $(\mathbf{A}, s)$ is any representative in $[\mathbf{A}, s]$. Clearly, F satisfies $F \circ \pi = \Phi$ and is hence a smooth injective function (here the smoothness of F is implied by for example [Lee(2013)] Theorem 4.29). By the chain rule, we have

$$d\Phi_{(\mathbf{A}, s)} = dF_{[\mathbf{A}, s]} \circ d\pi_{(\mathbf{A}, s)}. \quad (1.6)$$

It follows immediately that $\ker(d\Phi_{(\mathbf{A}, s)}) \subseteq \ker(d\pi_{(\mathbf{A}, s)})$. The other direction will follow if we can prove that $dF_{[\mathbf{A}, s]}$ is injective. To do so, suppose that $v \in T_{[\mathbf{A}, s]}M$ is such that $dF_{[\mathbf{A}, s]}(v) = 0$. Let $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$ be

the horizontal lift of v , meaning that $v = d\pi_{(\mathbf{A},s)}((\mathbf{V}, c))$. (1.6) then implies that $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A},s)})$ and the proceeding discussion shows that $c = 0$ and $\mathbf{V} = \mathbf{A}\Omega$ for some skew-symmetric Ω . To proceed, we look at the curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ defined by $\gamma(t) = (\mathbf{A}e^{t\Omega}, s)$. Clearly $\gamma(0) = (\mathbf{A}, s)$ and $\gamma'(0) = (\mathbf{A}\Omega, 0)$. Moreover, the matrix $e^{t\Omega}$ is always orthogonal for any t and any skew-symmetric Ω so that $\pi(\gamma(t)) = [\mathbf{A}, s]$ for all t . We have

$$v = d\pi_{(\mathbf{A},s)}((\mathbf{A}\Omega, 0)) = d\pi_{(\mathbf{A},s)}(\gamma'(0)) = \frac{d}{dt} \big|_{t=0} \pi(\gamma(t)) = \mathbf{0} \in T_{[\mathbf{A},s]}M.$$

Therefore, $dF_{[\mathbf{A},s]}$ is injective and the claim is proved. The verticle tangent space is therefore the same as $\ker(d\Phi_{(\mathbf{A},s)})$.

To compute the horizontal tangent space, any $(\mathbf{V}, c) \in V_{(\mathbf{A},s)}^\perp$ will satisfy

$$0 = g_{(\mathbf{A},s)}((\mathbf{V}, c), (\mathbf{A}\Omega, 0)) = \text{tr}(\mathbf{A}^T \mathbf{V} \Omega) \quad (1.7)$$

for all skew-symmetric Ω . Decomposing $\mathbf{A}^T \mathbf{V} = \mathbf{S} + \mathbf{K}$, where $\mathbf{S}$ is the symmetric part and $\mathbf{K}$ is the skew-symmetric part, we find out that

$$0 = \text{tr}(\mathbf{S}\Omega) + \text{tr}(\mathbf{K}\Omega) = 0 + \text{tr}(\mathbf{K}\Omega) = \text{tr}(\mathbf{K}\Omega).$$

Setting $\Omega = \mathbf{K}^T$ will yield that $\mathbf{K} = \mathbf{0}$. Therefore, $\mathbf{A}^T \mathbf{V}$ is a symmetric matrix. Conversely, any $\mathbf{V}$ such that $\mathbf{A}^T \mathbf{V}$ is symmetric will satisfy (1.7). The proof is complete. $\square$

Proposition 2. For every $[\mathbf{A}, s] \in M$ and every $v \in T_{[\mathbf{A},s]}M$

References

- [Lee(2013)] John M. Lee. *Introduction to Smooth Manifolds*, volume 218 of *Graduate Texts in Mathematics*. Springer New York, second edition, 2013. ISBN 978-1-4419-9981-8. doi: 10.1007/978-1-4419-9982-5.
- [Lee(2018)] John M. Lee. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer International Publishing, second edition, 2018. ISBN 978-3-319-91754-2. doi: 10.1007/978-3-319-91755-9.